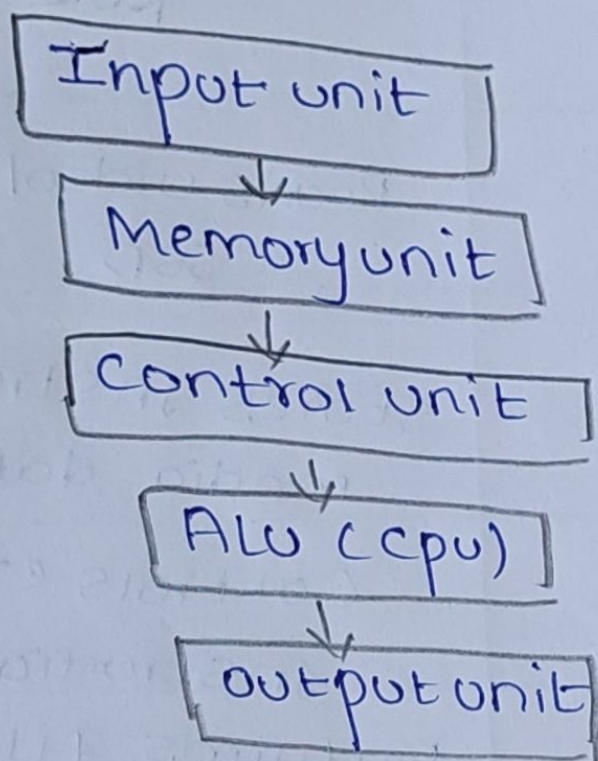


Functional Units of computer:

concept MAP:-



Explanation:-

The input unit reads video frames and audio data from SSD and loads them into RAM. The Memory unit stores both multimedia data and program instructions. The control unit fetches instructions from memory decodes them and directs the ALU to perform operations. The ALU applies filters, compresses the video synchronizes audio and processes image data. Finally, the processed output is sent to the

output unit, where it is displayed as preview
Saved to the storage device

Coordination of five functional units:-

Functional unit	Role
Input unit	Reads video/audio data from SSD
memory unit	stores instruction & multi media data
Control unit	Controls execution of instruction
ALU	Performs filtering compression and calculations
output unit	Displays preview & stores final video

2. Instruction execution cycle & performance Factors:

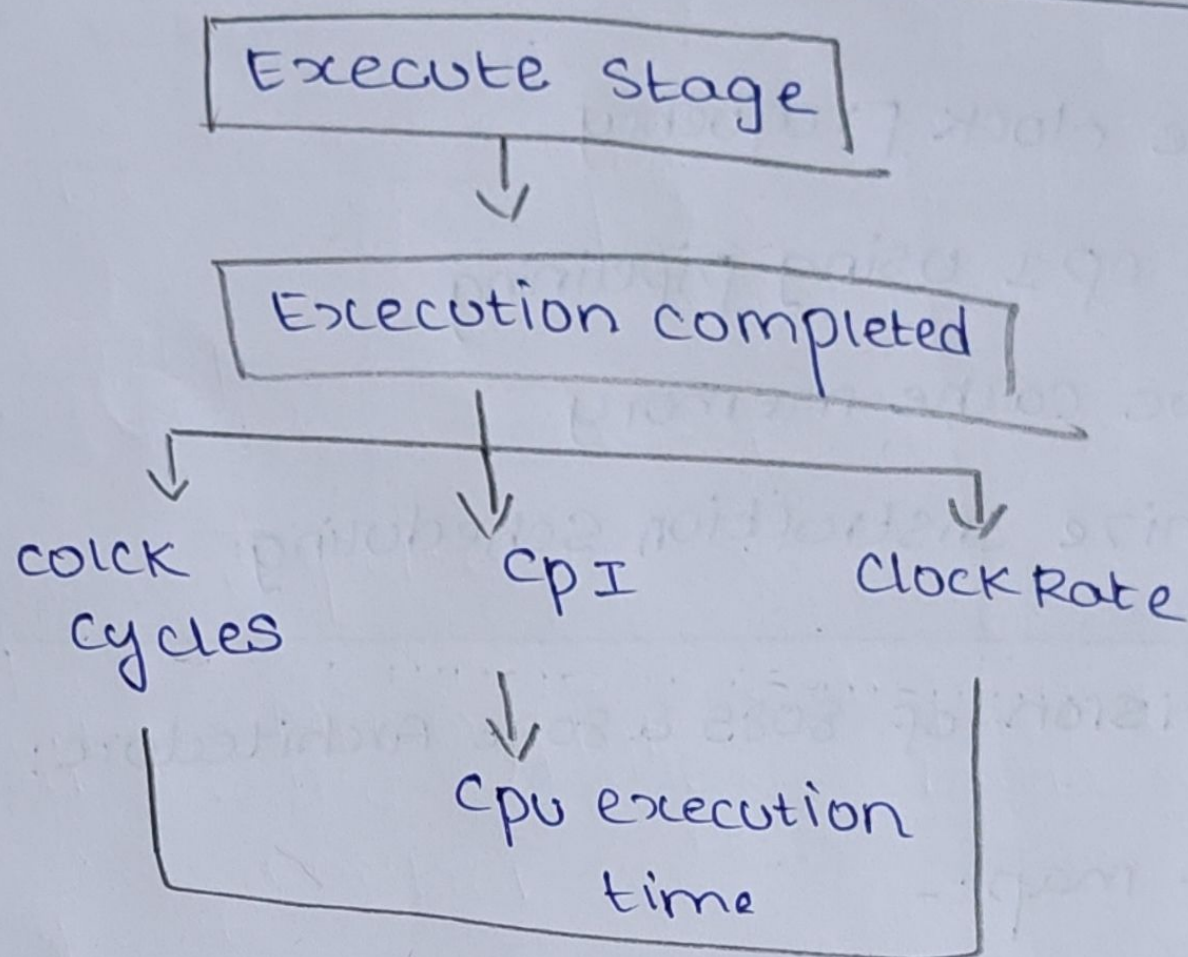
Program Instructions



Fetch Stage



Decode Stage



Formula: -

$$\text{CPU execution time} = \frac{\text{Instruction count} \times \text{CPI}}{\text{Clock Rate}}$$

Explanation: -

- Fetch retrieves instruction from memory
- Decode identifies the operation & operands
- Execute performs the operation using the ALU

Performance factors: -

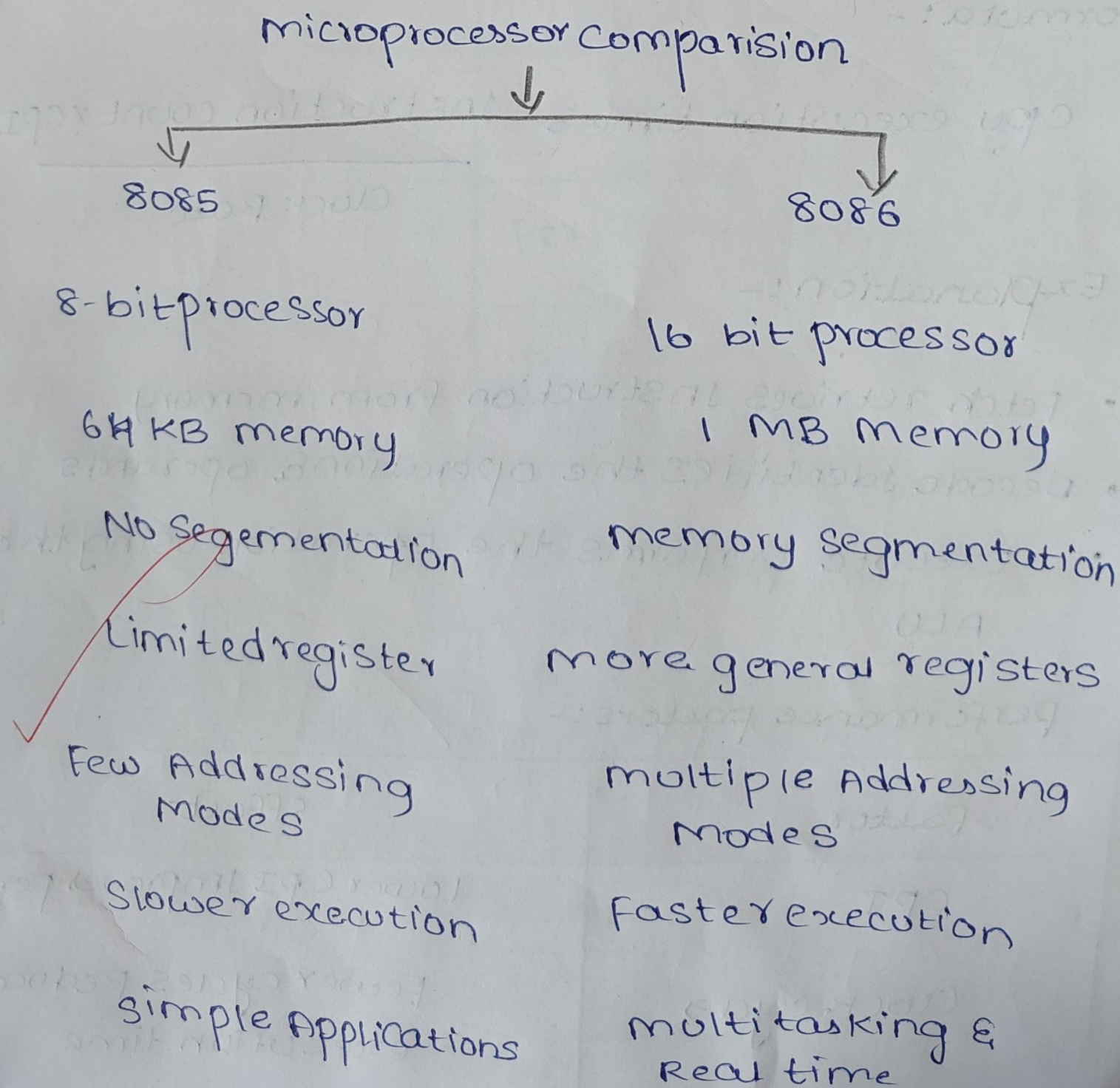
Factor	effect
CPI	Lower CPI Improves performance
Clock cycle	Fewer cycles Reduce execution time

Improvement Methods:

- Increase clock frequency
- Reduce CPI using pipelining
- Improve cache memory
- Optimize instruction scheduling

3. Comparision of 8085 & 8086 Architecture:

Concept map:-



Comparison table: -

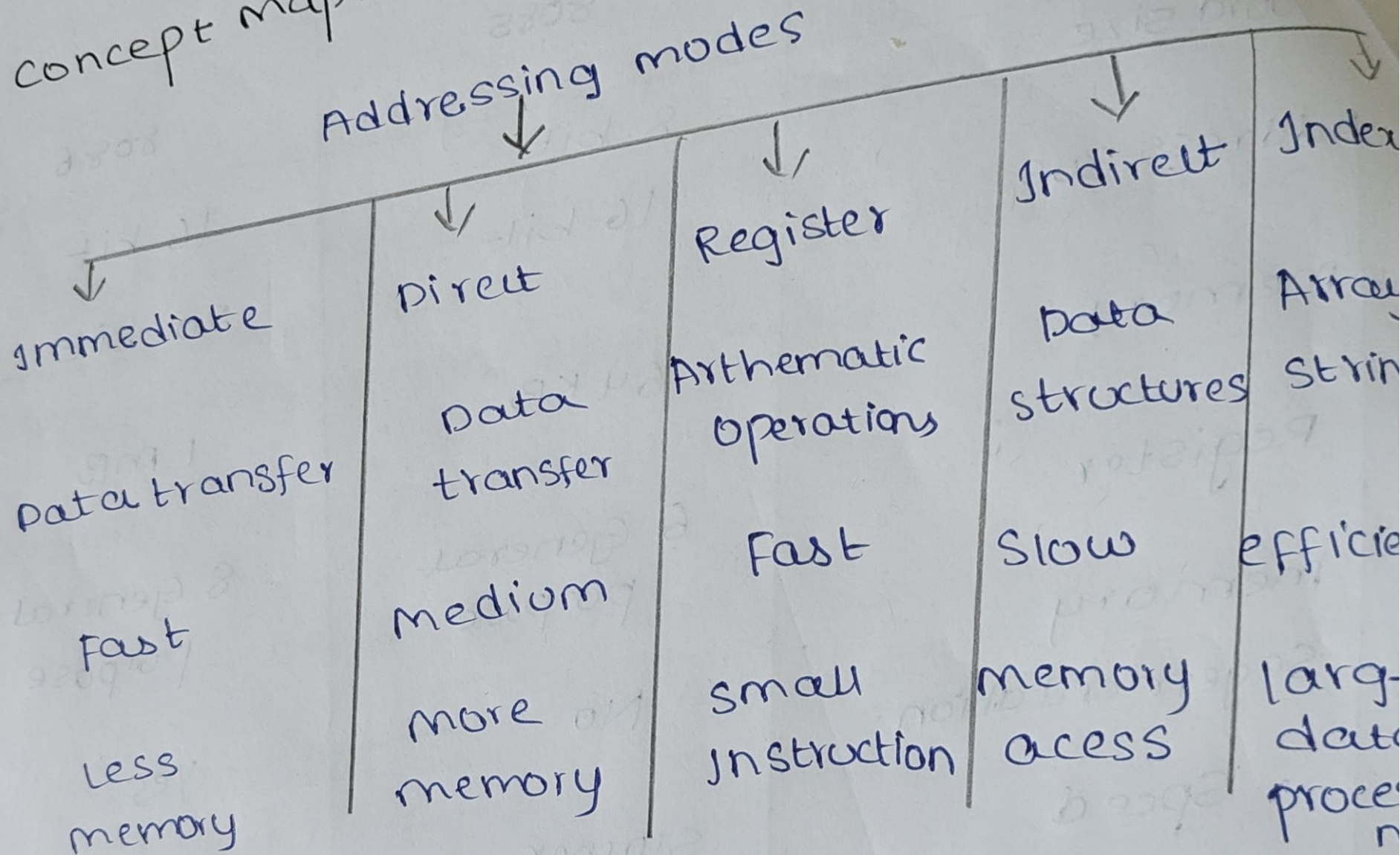
Feature	8085	8086
Word size	8-bit	16-bit
Address Bus	16-bit	20-bit
Memory	64 KB	1 MB
Register	6 General purpose	8 General purpose
Memory Segmentation	No	Yes
Speed	Slower	Faster
Multitasking	Limited	Better

Explanation :-

The 8085 is suitable for simple embedded systems because of its simple architecture. The 8086 offers higher processing speed, large memory, and better multitasking. It is suitable for robotics and real time systems.

Addressing modes

concept map:-



Applications:-

Addressing Mode	Application
Immediate	load constants
Direct	Access fixed memory locations
Register	Fast arithmetic calculations
Indirect	Stack and linked lists
indexed	Arrays, matrices, strings

Addressing :-

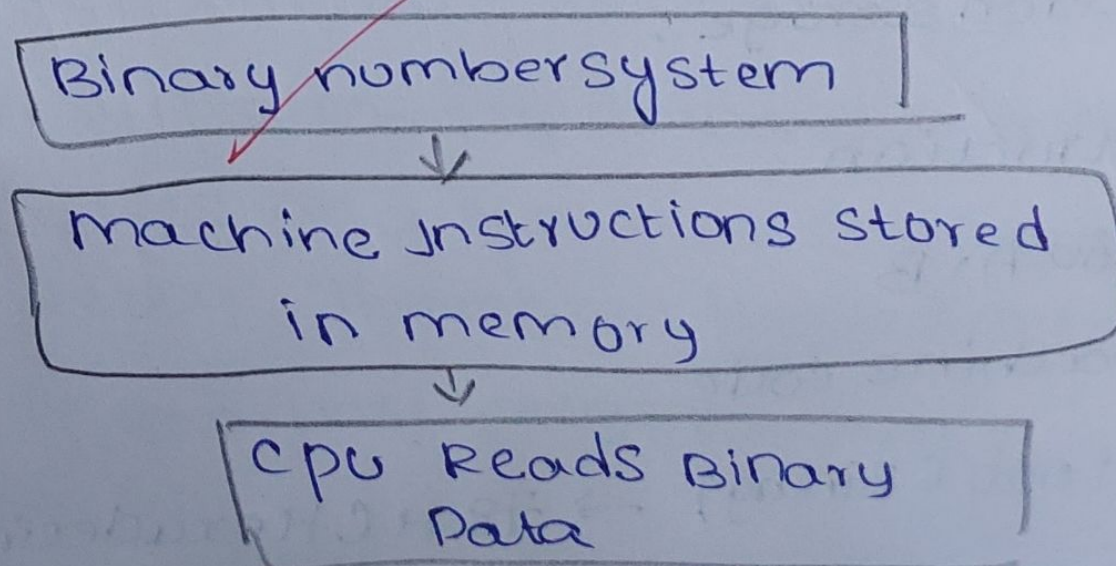
Addressing mode	Memory Access	Instruction size	Speed
Immediate	None	Small	Very fast
Register	No memory	Small	Fastest
Direct	One memory Access	medium	Moderate
Indirect	Multiple Accesses	medium	Slower
Indexed	Indexed Access	Larger	Efficient for arrays

Conclusion:-

Choosing the correct addressing mode reduces memory access time, improves execution speed and increases processor efficiency.

5. Binary & Hexadecimal number systems:

Concept map:-



Executes Instruction



Binary to Hexadecimal
Conversion



Hexadecimal conversion
Number System



Easier debugging &
programming

Binary to hexadecimal conversion:-

Example:

Binary : 10101111

Group into 4 bits

1010

1111

A

F

hexadecimal = AF

Instruction storage :

Instruction

mov A, B

machine code

0111000 (Binary) → 78 H (Hexadecimal)

Hexadecimal is preferred:-

Shorter representation

Easy to read

Easy debugging

One hexadecimal digit represents 4 binary bits

Convenient for memory addresses

Applications :-

Binary	Hexadecimal
cpu execution	debugging
memory storage	memory addresses
Digital logic	Assembly programming
Instruction execution	Machine code representation

Conclusion:-

Machine instructions are stored in binary, which the CPU executes directly. Hexadecimal is a compact representation of binary making memory addresses and machine code easier to read and debug. Therefore hexadecimal is widely used in low-level programming & system design.